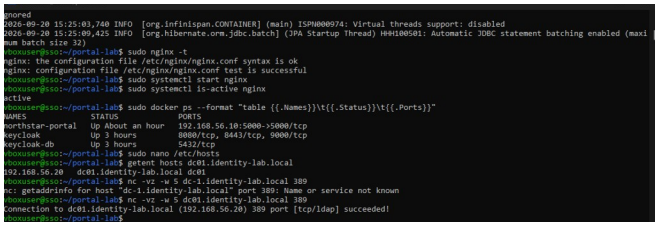
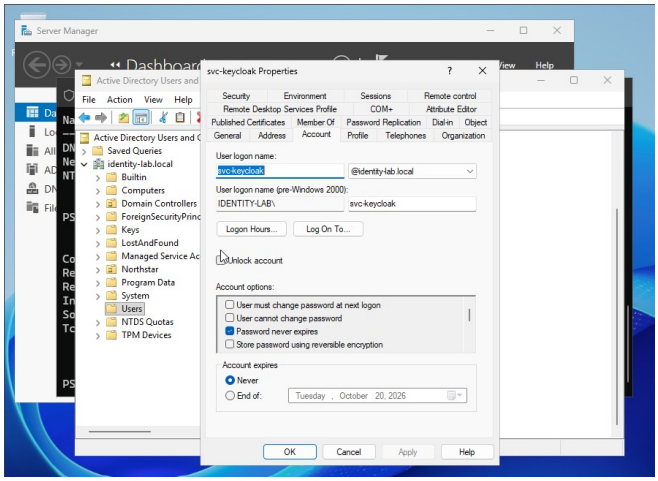
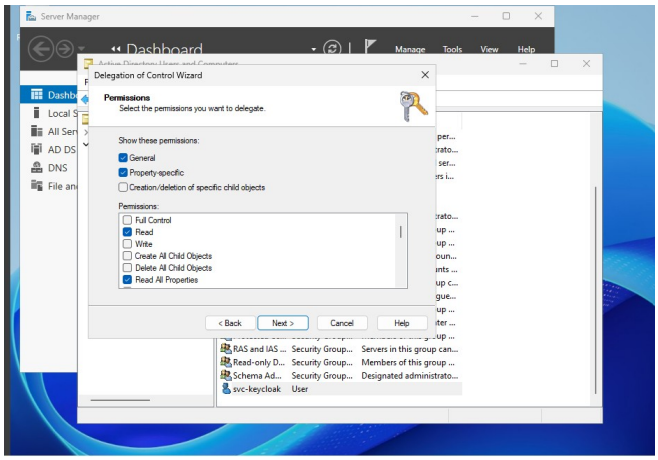
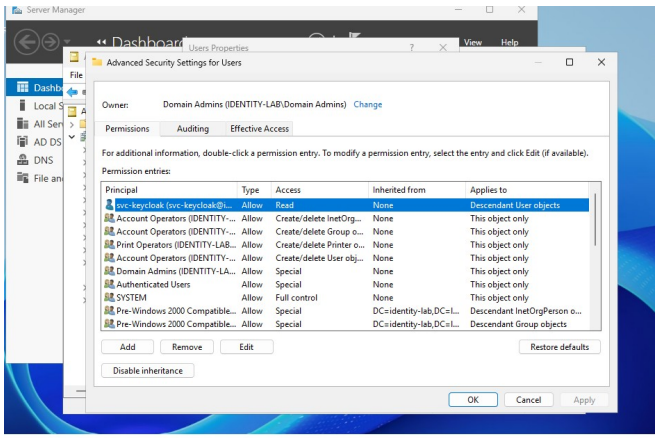
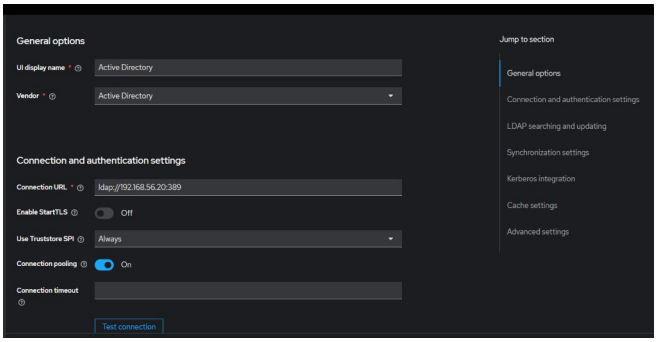
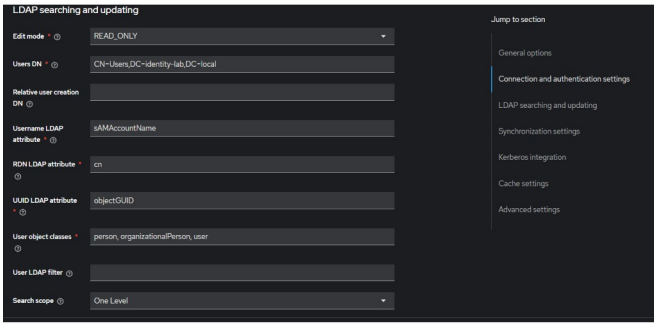
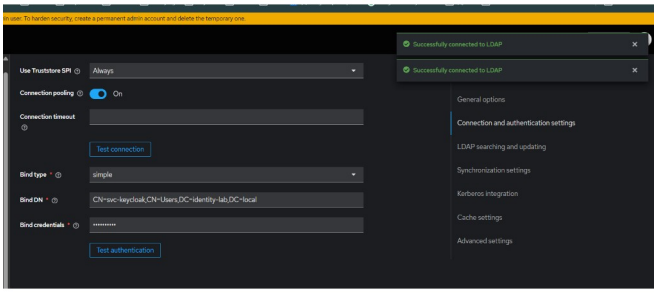
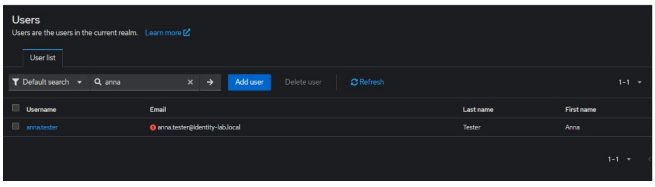
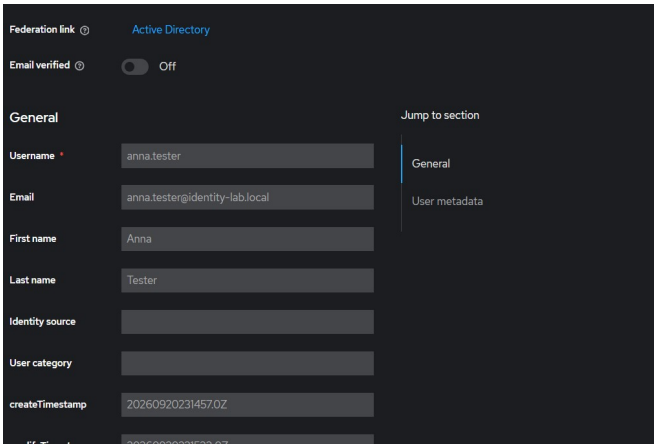
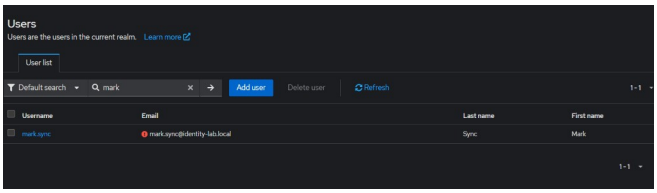
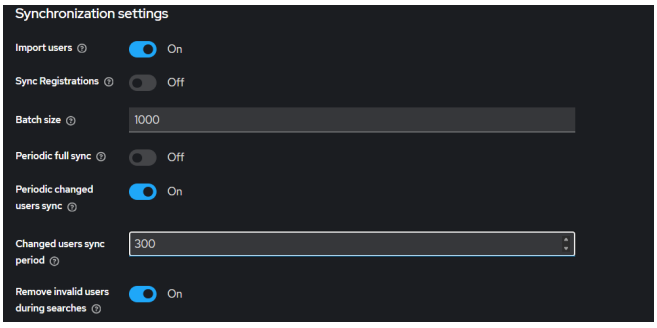
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| The domain controller’s hostname was mapped to its IP address in the Ubuntu hosts file. A connection test confirmed that LDAP port 389 was reachable. |  |
| The svc-keycloak service account was created in the Active Directory Users container. It was configured without requiring a password change at the next logon. |  |
| Read permissions for user objects were delegated to svc-keycloak using the Delegation of Control Wizard. No additional write, create, or delete permissions were granted. |  |
| The security settings of the Users container were inspected to verify the permissions assigned to svc-keycloak. The delegated read permissions applied to descendant user objects. |  |

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| An LDAP provider was configured in Keycloak with Active Directory selected as the vendor and ldap://192.168.56.20:389 as the connection URL. The svc-keycloak account was specified for directory authentication. |  |
| The LDAP provider was configured in READ_ONLY mode to search the Active Directory Users container. The sAMAccountName, cn, and objectGUID attributes were selected for username, RDN, and UUID mapping. |  |
| The Test connection operation completed successfully. This confirmed that Keycloak could establish a connection to the Active Directory LDAP service. |  |
| A full LDAP synchronization was performed to import users from Active Directory into Keycloak. The anna.tester account appeared in the realm's user list. |  |

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| The imported anna.tester profile was inspected to verify its username, email, and name attributes. The Federation link confirmed that the account was associated with the Active Directory LDAP provider. |  |
| The anna.tester user successfully signed in to the Keycloak Account Console using the password configured in Active Directory. This verified authentication through the LDAP federation. |  |
| The mark.sync account was created in Active Directory and appeared in Keycloak after LDAP synchronization. This confirmed that newly created directory users could be imported without manual account creation in Keycloak. |  |
| A dedicated Group Policy was configured and applied to allow unsigned LDAP binds in the isolated lab environment. LDAP signing enforcement was disabled, and the LDAP server signing requirement was set to None. |  |